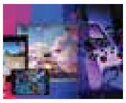
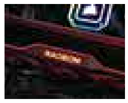


FB to expand Cloud Gaming
Facebook Cloud Gaming Sees 1.5 Mn Users A Month, Expanding To More Regions.
http://dgit.in/Jul21-67



Micro Center apologizes to AMD
US retailer Micro Center apologizes for insulting AMD graphics cards.
http://dgit.in/Jul21-68

Install Timeshift via APT as follows

```
sudo apt install timeshift
```

After a successful install, open Timeshift from the Activities menu on the top left, or by hitting the Super key. The first time, timeshift will ask for configuration. You may choose to expand the help dropdown for extra info. On the first screen, select btrfs. Hit Next. On the next screen, select the disk which has our root partition. It will be easy to identify, as it will have a size greater than the disk size. Since we added our second disk to the first filesystem, selecting one will also automatically select the other. Hit Next. In the next screen, you will be asked to setup a schedule for snapshots. You can tune this to your preferred settings. A sane starting point is 5 daily backups, 3 weekly backups, and 2 monthly backups. Its also a good idea to keep a few 'Boot' backups, which save a known working state of the computer 10 minutes after every time you start your computer. When you are done, click "Next".

In this section, enable both the options: to inlcude @home in the backup, and to enable BTRFS qgroups. The first option also backups your personal files and folders along with core system data. The second option gives you an accurate report of space used by each snapshot. Hit Next.

The final screen is a short conclusion of your settings and expected behaviour. Notice that the last point warns you that btrfs snapshots based backups will not save you from a disk failure, and on first glance suggest that you should use rsync based backups to avoid that. This is misleading, as rsync based backups on the same system will also not guard against disk failures. To move your btrfs based backups onto an external drive, you can use btrfs-send and btrfs-receive utilities.

Click on Finish to proceed to the main program interface. In this interface, you can hit "Create" to make your first snapshot. You can see how quick and easy it is to backup files via btrfs! Note how in the screenshot I have two backups of nearly 6 GB worth of data each, but their actual sizes are just 1.3 MB and 163.8 KB!

> WARNING: It is possible that when you fiddle about at Settings and return to the main window, it will warn you that you are out of space and backups will no longer be created. This is in error, and is because it is checking the wrong device, as our devices are pooled together via btrfs. Simply close and open Timeshift again to fix this.

timeshift-autosnap-apt

timeshift-autosnap-apt is an optional tool you can install in this setup. This tool takes a snapshot backup of your system every time the APT package manager is invoked : so, before install, uninstall, or update operation. This is a convenient way to have automatic backups in case an update or a new package break your system. First, we install the dependencies for the autosnap utility.

```
sudo apt install git make
```

Then, we clone the git repo to your computer, and compile the utility.

```
mkdir -p ~/.deb
git clone https://github.com/wmutschl/timeshift-autosnap-apt ~/.deb/timeshift-autosnap-apt
cd ~/.deb/timeshift-autosnap-apt
sudo make install
```

Once installed, we tune the configuration file to our setup by running sudo nano /etc/timeshift-autosnap-apt.conf

We set snapshotBoot to false, maxSnapshots to 5, and updateGrub to false. Once your config file is tuned to your liking, save and exit nano. Your computer will now backup before any package changes done via apt!

DATA DEDUPLICATION

On btrfs, one can also deduplicate data. Deduplication here doesn't change the number of files you end up with, but saves on space by linking identical data, so its stored only once, and all the copies merely need to link to the original copy. Deduplication on btrfs happens on a filesystem block level, not just on a file level. So, if two files are not identical, but share some blocks of data, you still save space using the same strategy.

We use the tool duperemove for the process. The utility reads all of your files to compute checksums of each block of each file, and then submits them to the Linux kernel as candidates for deduplication. This is a safe method, so that none of your data is lost in the process.

Install dupermove from apt. Note that if you have timeshift-autosnap-apt installed, you might see much more output than you are used to. This is because the autosnapper will take a system backup after you are done downloading duperemove, but before installing it. In case you already have 5 autosnap backups, it will also delete the oldest backup. All this extra activity is printed to your terminal.

```
sudo apt install duperemove
```

Once you are done downloading, you can run duperemove. Its a good idea to let duperemove maintain a list of file checksums so that subsequent runs of the application are incremental, and thus, faster. You can add the invocation to a crontab so it runs on its own every once in a while, or run it manually when you think you have enough data duplicates that its turning into a problem.

```
sudo duperemove -dhr --hashfile=/.dedupehash /
```

This process may take a while, but will display a summary when its done. On even just a fresh install, I was able to save 28 MB!